

Lightweight and Reproducible Cassava Leaf Disease Classification

A Controlled Study of Architecture and Training Design on Field Images

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1. Why this Problem Matters

- Cassava is a critical food-security crop in Africa
- More than 70 million people in Africa depend on cassava
- At least 80% of smallholder farmer households in Sub-Saharan Africa grow cassava
- Disease threatens yields
- Diagnosis from mobile images is a realistic use case, so lightweight/reproducible systems matter for practical field deployment

2. Dataset and Controlled Evaluation Setup

- Dataset contains 21,397 validated and labeled field images (Size 800 by 600 RGB)
- Supervised 5-class classification
- Classes: CBB, CBDSD, CGM, CMD, Healthy (Imbalance present, dominated by CMD)
- Train/Validation/Test split is a frozen 70/15/15 stratified split
- Primary selection metric is validation Macro F1 due to class imbalance

3. Controlled Study Design

Experiment 1: Architecture Ablation

- Baseline CNN
- Residual CNN
- Residual Attention CNN

Experiment 2: Training Strategy Ablation

- Setup A: Basic image augmentation
- Setup B: Stronger, field-oriented image augmentation
- Setup C: Setup B and weighted cross-entropy loss

Only one factor changes at a time so comparisons remain interpretable

4. Key Results

Experiment 1	Validation Macro F1
Baseline CNN	0.6776
Residual CNN	0.7107
Residual Attention CNN	0.7260

Residual Attention CNN won Experiment 1, so it was carried over to Experiment 2. In Experiment 2, the simplest training setup, A, won.

Experiment 2	Validation Macro F1
Setup A	0.7449
Setup B	0.7345
Setup C	0.7333

Final Held-Out Test Results	
Selected System	Test Metrics
Residual Attention CNN	Acc 0.8452
	F1 0.7317
	Loss 0.4918
Setup A	Acc 0.8589
	F1 0.7365
	Loss 0.4183

5. Complexity vs. Performance Tradeoff

Experiment 1: Complexity from Increasing # Parameters
Baseline CNN: 390,181 params
Residual CNN: 1,227,685 params (+837,504; +214.6%; 3.15x baseline) Residual vs. Baseline Macro F1 gain: +0.0331 (+4.9%)
Residual Attention CNN: 1,239,981 params (+12,296; +1.0% vs residual) Residual attention vs. Residual Macro F1 gain: +0.0153 (+2.2%)

Residual learning gave the biggest gain but at a large size cost. Lightweight attention added only about 1% more parameters beyond the residual model and still improved Macro F1, making it a comparatively parameter-efficient refinement.

Experiment 2: Complexity from Image Augmentation Training Setups
Setup B vs. A Macro F1: -0.0104 (approx. -1.4%)
Setup C vs. A Macro F1: -0.0116 (approx. -1.6%)

More complex training was not justified; the simplest setup generalized best.

6. Class-Level Behavior and Deployment Relevance

Final Winning System (Residual Attention CNN + Setup A) Test Recall

CBB		0.509
CBSD		0.729
CGM		0.732
CMD		0.970
Healthy		0.668

Confusion-Matrix Takeaway: Some CBB cases are confused with Healthy, and some classes remain visually overlapping under realistic field conditions. Strong average metrics therefore do not guarantee equally reliable disease recognition for every category.

Strongest class: CMD (0.970)
Weakest class: CBB (0.509)
Performance remained uneven across classes.

7. Core Insights

I. A controlled lightweight study was informative

Holding most of the pipeline fixed made it possible to isolate certain design choices, thereby providing insight into marginal performance improvements.

II. Architectural Complexity helped selectively, but training complexity did not

Residual connections improved over the baseline, lightweight attention improved further at very low marginal parameter cost, and the simplest training setup generalized best.

III. Field-image deployment still requires class-level caution

Overall results were strong, but reliability remained uneven across classes, especially between CMD and CBB.

Overall Contribution: Practical understanding of the tradeoff among predictive performance, efficiency, reproducibility, and deployability.